

Underdetermined DOA and Polarization Estimation for Arbitrary Polarization Sensitive Arrays

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Abstract—In this paper, direction of arrival (DOA) and polarization estimation in the underdetermined scenario are studied, where the number of sources exceeds the number of antennas in a polarization sensitive array. A virtual array model based on polarization sensitive arrays (PSAs) with arbitrary structure is derived by decoupling DOA-dependent and polarization-dependent parts of the vectorized covariance matrix. Subsequently, the sparse representation framework is established, and DOA estimation is performed by solving an optimization problem with additional constraints arising from the virtual array model. Polarization parameters are then estimated based on the DOA estimation results and least-square-based method. Simulation results demonstrate the superior performance of the proposed method compared with existing methods in the underdetermined scenario.

Index Terms—DOA estimation, polarization parameter estimation, underdetermined, sparse representation, polarization sensitive array.

I. INTRODUCTION

Direction of arrival (DOA) estimation has been widely studied in the past decade [1]–[8]. Based on antenna arrays, DOA estimation can be applied to various fields such as radar, navigation, and wireless communications. Polarization sensitive arrays (PSAs) are composed of antennas with diverse types and orientations, enabling the estimation of polarization parameters which can provide additional information for applications such as interference suppression and target identification.

Various methods have been proposed for DOA estimation with traditional scalar arrays (SAs), such as the MUSIC method [1], the ESPRIT method [2], and the sparse-representation-based (SR-based) methods [3]–[5]. For PSAs, extensions of the MUSIC [6] and ESPRIT methods [9] have been developed, as well as the SR-based methods [7], [10]. However, these methods are limited to the overdetermined scenario, where the number of sources is less than the number of antennas.

For DOA estimation in the underdetermined scenario, the concept of difference co-array is proposed, generated by vectorizing the covariance matrix of the array output [11]. Utilizing the concept, a virtual array with more virtual sensors is obtained from a given physical array. Following the nested array [11], a variety of sparse array configurations have been proposed, such as the coprime array [12], [13], the super

nested array [14], and the fractal array [15], [16]. For underdetermined DOA estimation, spatial-smoothing-based MUSIC (SS-MUSIC) [17] and compressive-sensing-based methods [18] can be applied.

Numerous estimation methods have also been proposed in recent years for PSAs in the underdetermined scenario. Most of them treat multiple antennas sharing the same location as a single vector sensor, enabling underdetermined DOA and polarization estimation when the number of sources exceeds the number of vector sensors. The cogenerated orthogonal dipole triad (CODT) composed of three dipoles is treated as a vector sensor in [19]. Underdetermined DOA and polarization estimation with a parallel coprime array consisting of CODTs is performed using integrating forward/backward spatial smoothing MUSIC with sparse representation. The cogenerated orthogonal dipole pair (COPD) with two orthogonal dipoles is considered as a single sensor in [20], and underdetermined parameter estimation is obtained by solving the optimization problem based on annihilating relation based on a coprime array composed of CODPs. In [21], the spatially separated loop and dipole pair (SSLDP) composed of a dipole and a loop is considered as a single sensor, with the submatrix fitting-based decoupling rooting rank deficiency method proposed for underdetermined parameter estimation. In [22], each antenna is regarded as a single sensor, and a synthetic nested array is proposed, along with a two-step method for estimation of parameters, where the source number is larger than the number of antennas. However, two-step method is exclusively applicable to synthetic nested array configuration.

In this work, we propose a method which can be applied to PSAs with arbitrary structures and suitable for underdetermined DOA and polarization parameter estimation. A virtual array model for an arbitrary PSA is constructed by decoupling the DOA-dependent and polarization-dependent parts in the vectorized covariance matrix. Subsequently, the sparse representation framework is employed, where DOA estimation is performed by solving a constraint optimization problem, followed by polarization parameter estimation using the least-square-based method.

The rest of this paper is organized as follows. The model formulation is given in Section II. Virtual array model and the proposed method are presented in Section III. Simulations are provided in Section IV and Section V concludes the paper.

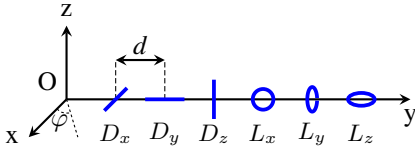


Fig. 1: The structure of a spatially separated EMVS.

II. MODEL FORMULATION

Consider a polarization sensitive linear array composed of N antennas in the Cartesian coordinate system, each antenna positioned at the y -axis with coordinate $\{y_i\}_{i=1}^N$. A simple example is provided in Fig. 1, where the spatially separated electromagnetic vector sensor (EMVS) composed of three orthogonal dipoles and three orthogonal loops are given.

K fully-polarized far-field narrowband zero-mean stationary signals impinge on the array with λ as the signal wavelength. The DOA, polarization angle, and polarization phase difference of the k -th ($1 \leq k \leq K$) signal are defined as φ_k , γ_k , and η_k , respectively.

The manifold of the k -th signal can be represented as

$$\mathbf{a}(\varphi_k, \gamma_k, \eta_k) = \text{diag}(\mathbf{p}(\varphi_k)) \mathbf{G} \Phi(\varphi_k, \gamma_k, \eta_k), \quad (1)$$

where $\mathbf{p}(\varphi_k) = [e^{j2\pi y_1 \sin \varphi_k / \lambda}, \dots, e^{j2\pi y_N \sin \varphi_k / \lambda}]^T$ is the spatial steering vector. $\mathbf{G} = [\mathbf{g}_1^T, \dots, \mathbf{g}_N^T]^T$ indicates the reception of electric and magnetic fields at the i -th antenna with $\mathbf{g}_i \in \mathbb{R}^{1 \times 6}$, which is determined solely by the antenna type and orientation. $\Phi(\varphi_k, \gamma_k, \eta_k)$ is defined as [9]

$$\Phi(\varphi_k, \gamma_k, \eta_k) = \mathbf{H}(\varphi_k) \boldsymbol{\beta}_k, \quad (2)$$

where

$$\mathbf{H}(\varphi_k) = \begin{bmatrix} -\sin \varphi_k & 0 \\ \cos \varphi_k & 0 \\ 0 & -1 \\ 0 & -\sin \varphi_k \\ 0 & \cos \varphi_k \\ 1 & 0 \end{bmatrix}, \quad \boldsymbol{\beta}_k = \begin{bmatrix} \cos \gamma_k \\ \sin \gamma_k e^{j\eta_k} \end{bmatrix}. \quad (3)$$

Define $\mathbf{B}(\varphi_k) = \text{diag}(\mathbf{p}(\varphi_k)) \mathbf{G} \mathbf{H}(\varphi_k)$, and the manifold can be decomposed as

$$\mathbf{a}(\varphi_k, \gamma_k, \eta_k) = \mathbf{B}(\varphi_k) \boldsymbol{\beta}_k, \quad (4)$$

where $\mathbf{B}(\varphi_k) = [\mathbf{b}_1(\varphi_k), \mathbf{b}_2(\varphi_k)]$ and $\boldsymbol{\beta}_k$ are the DOA-dependent part and the polarization-dependent part, respectively.

Assuming the K signals are statistically uncorrelated and corrupted by zero-mean additive white Gaussian noise, the array output can be formulated as

$$\mathbf{x}(t) = \mathbf{A} \mathbf{s}(t) + \mathbf{n}(t), \quad (5)$$

where $\mathbf{x}(t) = [x_1(t), \dots, x_N(t)]^T$ represents the received signal vector, $\mathbf{A} = [\mathbf{a}(\varphi_1, \gamma_1, \eta_1), \dots, \mathbf{a}(\varphi_K, \gamma_K, \eta_K)]$ represents the steering matrix, $\mathbf{s}(t) = [s_1(t), \dots, s_K(t)]^T$ represents the source signal vector, and $\mathbf{n}(t) = [n_1(t), \dots, n_N(t)]^T$ represents the noise vector.

III. PROPOSED METHOD

In this section, we present a virtual array model based on vectorized PSA covariance matrix by decoupling DOA-dependent and polarization-dependent parts. Subsequently, the sparse representation framework is established for an arbitrary PSA structure. Building upon this, DOA and polarization estimation method are proposed utilizing the concept of group sparsity and least-square-based method.

A. Virtual Array Model for Arbitrary PSAs

The covariance matrix of $\mathbf{x}(t)$ is obtained by

$$\mathbf{R} = \mathbb{E}\{\mathbf{x}(t)\mathbf{x}^H(t)\} = \mathbf{A} \mathbf{R}_s \mathbf{A}^H + \sigma_n^2 \mathbf{I}_N, \quad (6)$$

where $\mathbf{R}_s = \text{diag}(\sigma_1^2, \dots, \sigma_K^2)$ is a diagonal matrix with the input arguments being its diagonal vector, due to the uncorrelation of the K signals, and $\sigma_k^2 = \mathbb{E}\{s_k(t)s_k^H(t)\}$ denotes the power of the k -th signal. $\mathbb{E}\{\cdot\}$ is the expectation operator, $\sigma_n^2 = \mathbb{E}\{n_i(t)n_i^H(t)\}$ is the noise power, and $\mathbf{I}_N$ denotes the N -dimensional identity matrix.

Considering the diagonal structure of $\mathbf{R}_s$, the covariance matrix can be rewritten as

$$\mathbf{R} = \sum_{k=1}^K \sigma_k^2 \mathbf{a}(\varphi_k, \gamma_k, \eta_k) \mathbf{a}^H(\varphi_k, \gamma_k, \eta_k) + \sigma_n^2 \mathbf{I}_N. \quad (7)$$

Vectorizing the covariance matrix and decoupling DOA-dependent and polarization-dependent parts, the vectorized covariance matrix can be written as

$$\begin{aligned} \mathbf{r} &= (\mathbf{A}^* \odot \mathbf{A}) \text{vec}(\mathbf{R}_s) + \sigma_n^2 \text{vec}(\mathbf{I}_N) \\ &= \sum_{k=1}^K \sigma_k^2 \mathbf{a}^*(\varphi_k, \gamma_k, \eta_k) \otimes \mathbf{a}(\varphi_k, \gamma_k, \eta_k) + \sigma_n^2 \vec{\mathbf{I}}_N \\ &= \sum_{k=1}^K \sigma_k^2 [\mathbf{B}^*(\varphi_k) \boldsymbol{\beta}_k^*] \otimes [\mathbf{B}(\varphi_k) \boldsymbol{\beta}_k] + \sigma_n^2 \vec{\mathbf{I}}_N, \end{aligned} \quad (8)$$

where $\text{vec}(\cdot)$, $\odot$, and $\otimes$ denote the vectorization operator, the Khatri-Rao product, and the Kronecker product, respectively.

According to the property $(\mathbf{A}\mathbf{B}) \otimes (\mathbf{C}\mathbf{D}) = (\mathbf{A} \otimes \mathbf{C})(\mathbf{B} \otimes \mathbf{D})$, we have

$$\begin{aligned} \mathbf{r} &= \sum_{k=1}^K \sigma_k^2 [\mathbf{B}^*(\varphi_k) \otimes \mathbf{B}(\varphi_k)] [\boldsymbol{\beta}_k^* \otimes \boldsymbol{\beta}_k] + \sigma_n^2 \vec{\mathbf{I}}_N \\ &= \sum_{k=1}^K [\bar{\mathbf{b}}_1(\varphi_k), \bar{\mathbf{b}}_2(\varphi_k), \bar{\mathbf{b}}_3(\varphi_k), \bar{\mathbf{b}}_4(\varphi_k)] \begin{bmatrix} \bar{\beta}_{1,k} \\ \bar{\beta}_{2,k} \\ \bar{\beta}_{3,k} \\ \bar{\beta}_{4,k} \end{bmatrix} + \sigma_n^2 \vec{\mathbf{I}}_N. \end{aligned} \quad (9)$$

Based on the definition of $\mathbf{B}(\varphi_k)$ and $\boldsymbol{\beta}_k$, $\bar{\mathbf{b}}_m(\varphi_k)$ can be further detailed as

$$\begin{aligned} \bar{\mathbf{b}}_1(\varphi_k) &= \mathbf{b}_1^*(\varphi_k) \otimes \mathbf{b}_1(\varphi_k), \\ \bar{\mathbf{b}}_2(\varphi_k) &= \mathbf{b}_1^*(\varphi_k) \otimes \mathbf{b}_2(\varphi_k), \\ \bar{\mathbf{b}}_3(\varphi_k) &= \mathbf{b}_2^*(\varphi_k) \otimes \mathbf{b}_1(\varphi_k), \\ \bar{\mathbf{b}}_4(\varphi_k) &= \mathbf{b}_2^*(\varphi_k) \otimes \mathbf{b}_2(\varphi_k), \end{aligned} \quad (10)$$

and

$$\begin{aligned}\bar{\beta}_{1,k} &= \sigma_k^2 \cos^2 \gamma_k, \\ \bar{\beta}_{2,k} &= \sigma_k^2 \cos \gamma_k \sin \gamma_k e^{j\eta_k}, \\ \bar{\beta}_{3,k} &= \sigma_k^2 \cos \gamma_k \sin \gamma_k e^{-j\eta_k}, \\ \bar{\beta}_{4,k} &= \sigma_k^2 \sin^2 \gamma_k.\end{aligned}\quad (11)$$

Combine all the above derivations, (9) can be rewritten as

$$\mathbf{r} = \bar{\mathbf{B}}\bar{\boldsymbol{\beta}} + \sigma_n^2 \bar{\mathbf{I}}_N, \quad (12)$$

where $\bar{\mathbf{B}} = [\bar{\mathbf{B}}_1, \bar{\mathbf{B}}_2, \bar{\mathbf{B}}_3, \bar{\mathbf{B}}_4]$, $\bar{\boldsymbol{\beta}} = [\bar{\boldsymbol{\beta}}_1^T, \bar{\boldsymbol{\beta}}_2^T, \bar{\boldsymbol{\beta}}_3^T, \bar{\boldsymbol{\beta}}_4^T]^T$.

Furthermore,

$$\begin{aligned}\bar{\mathbf{B}}_m &= [\bar{\mathbf{b}}_m(\varphi_1), \dots, \bar{\mathbf{b}}_m(\varphi_K)], \\ \bar{\boldsymbol{\beta}}_m &= [\bar{\beta}_{m,1}, \dots, \bar{\beta}_{m,K}]^T, \quad m = 1, 2, 3, 4.\end{aligned}\quad (13)$$

From the perspective of virtual array generation, it is evident that $\mathbf{r}$ is the virtual array output, $\bar{\mathbf{B}}$ represents the $N^2 \times 4K$ virtual array steering matrix, and $\bar{\boldsymbol{\beta}}$ represents the virtual source signal vector depending on the polarization and power of each signal.

B. Underdetermined DOA and Polarization Parameter Estimation Method

Defining a search grid of K_g ($K_g \gg K$) potential DOAs, represented as $(\varphi_{g,1}, \dots, \varphi_{g,K_g})$, and the dictionary matrix can be represented as

$$\bar{\mathbf{B}}_g = [\bar{\mathbf{B}}_{g,1}, \bar{\mathbf{B}}_{g,2}, \bar{\mathbf{B}}_{g,3}, \bar{\mathbf{B}}_{g,4}], \quad (14)$$

where

$$\bar{\mathbf{B}}_{g,m} = [\bar{\mathbf{b}}_m(\varphi_{g,1}), \bar{\mathbf{b}}_m(\varphi_{g,2}), \dots, \bar{\mathbf{b}}_m(\varphi_{g,K_g})]. \quad (15)$$

Under the sparse reconstruction framework, (12) can be represented as

$$\mathbf{r} = \bar{\mathbf{B}}_g \boldsymbol{\psi} + \sigma_n^2 \bar{\mathbf{I}}_N, \quad (16)$$

where $\boldsymbol{\psi} = [\psi_1^T, \psi_2^T, \psi_3^T, \psi_4^T]^T$.

Clearly, ψ_m ($m = 1, 2, 3, 4$) are sparse, each containing K non-zero elements whose values are $\bar{\beta}_{m,k}$ ($1 \leq k \leq K$), respectively. Furthermore, define $\boldsymbol{\Psi} = [\psi_1, \psi_2, \psi_3, \psi_4]$ and the four column vectors of $\boldsymbol{\Psi}$ share the same spatial support.

Based on the definition of $\bar{\beta}_{m,k}$ in (11), it is apparent that $\bar{\beta}_{1,k}$ and $\bar{\beta}_{4,k}$ are non-negative real numbers, while $\bar{\beta}_{2,k}$ and $\bar{\beta}_{3,k}$ are conjugate complex numbers. Therefore, the constraint optimization problem for DOA estimation utilizing the group sparsity concept is formulated as

$$\begin{aligned}\min_{\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3, \mathbf{s}_4, \sigma_n^2} \quad & \|\boldsymbol{\Psi}\|_{2,1} \\ \text{subject to} \quad & \|\mathbf{r} - \bar{\mathbf{B}}_g \boldsymbol{\psi} - \sigma_n^2 \bar{\mathbf{I}}_N\|_F \leq \epsilon \\ & \psi_1 = \mathbf{s}_1, \\ & \psi_2 = \mathbf{s}_2 + j\mathbf{s}_3, \\ & \psi_3 = \mathbf{s}_2 - j\mathbf{s}_3, \\ & \psi_4 = \mathbf{s}_4, \\ & \mathbf{s}_1, \mathbf{s}_4 \in \mathbb{R}_+^{K_g \times 1}, \\ & \mathbf{s}_2, \mathbf{s}_3 \in \mathbb{R}^{K_g \times 1}, \\ & \sigma_n^2 \in \mathbb{R}_+, \end{aligned}\quad (17)$$

where $\|\cdot\|_{2,1}$ and $\|\cdot\|_F$ represent the mixed $\ell_{2,1}$ norm and the Frobenius norm, respectively. $\mathbb{R}^{P \times Q}$ and $\mathbb{R}_+^{P \times Q}$ denote the $P \times Q$ real matrices and non-negative real matrices.

The optimization problem can be solved using CVX software package [23], and the DOA estimation results, obtained by identifying the non-zero rows of $\boldsymbol{\Psi}$, are defined as $\hat{\varphi}_k$ ($1 \leq k \leq K$).

The steering matrix of virtual array can be constructed based on DOA estimation results as

$$\hat{\mathbf{B}} = [\hat{\mathbf{B}}_1, \hat{\mathbf{B}}_2, \hat{\mathbf{B}}_3, \hat{\mathbf{B}}_4], \quad (18)$$

where

$$\hat{\mathbf{B}}_m = [\bar{\mathbf{b}}_m(\hat{\varphi}_1), \dots, \bar{\mathbf{b}}_m(\hat{\varphi}_K)], \quad m = 1, 2, 3, 4. \quad (19)$$

Solving (12) based on $\hat{\mathbf{B}}$ and the least-square-based method, $\bar{\boldsymbol{\beta}}$ is estimated by

$$\begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\sigma}_n^2 \end{bmatrix} = \left(\begin{bmatrix} \hat{\mathbf{B}}, \bar{\mathbf{I}}_N \end{bmatrix}^H \begin{bmatrix} \hat{\mathbf{B}}, \bar{\mathbf{I}}_N \end{bmatrix} \right)^{-1} \begin{bmatrix} \hat{\mathbf{B}}, \bar{\mathbf{I}}_N \end{bmatrix}^H \mathbf{r}. \quad (20)$$

Then, the polarization parameters are

$$\begin{aligned}\hat{\gamma} &= \text{atan}(\sqrt{\hat{\beta}_4/\hat{\beta}_1}), \\ \hat{\eta} &= (\angle \hat{\beta}_2 - \angle \hat{\beta}_3)/2,\end{aligned}\quad (21)$$

where $\hat{\boldsymbol{\gamma}} = [\hat{\gamma}_1, \hat{\gamma}_2, \dots, \hat{\gamma}_K]^T$, $\hat{\boldsymbol{\eta}} = [\hat{\eta}_1, \hat{\eta}_2, \dots, \hat{\eta}_K]^T$ are automatically paired with the DOA estimation results $\hat{\varphi}$.

IV. SIMULATION RESULTS

Consider a synthetic nested array (SNA) [22] composed of 5 z -dipoles and 5 z -loops. Therefore, there are 10 antennas in total. The array structure is shown in Fig. 2, and comparative simulations are conducted between the proposed method and two-step method [22].

A. Resolution Capacity

Firstly, the number of sources is set to 12, exceeding the number of antennas and thereby creating an underdetermined scenario. Assume that the sources are uniformly distributed within $\{(\varphi, \gamma, \eta) \mid \varphi \in [-35^\circ, 25^\circ], \gamma \in [15^\circ, 55^\circ], \eta \in [30^\circ, 60^\circ]\}$. The signal-to-noise ratio (SNR) is set to 15dB and the number of snapshots is 4000, and the spatial spectrum of the two methods are shown in Fig. 3, where the actual DOAs are indicated by black lines. It is clear that both the proposed method and the two-step method is capable of resolving all 12 sources.

Subsequently, the number of sources is increased to 15 with the signal distribution, SNR, and the number of snapshots unchanged. The spatial spectrum of the two methods are shown in Fig. 4. Although both methods perform well in the scenario with 12 sources, only the proposed method succeed to distinguish all the 15 sources in Fig. 4. As a result, increased degrees of freedom can be exploited by the proposed method, leading to improved resolution capacity.

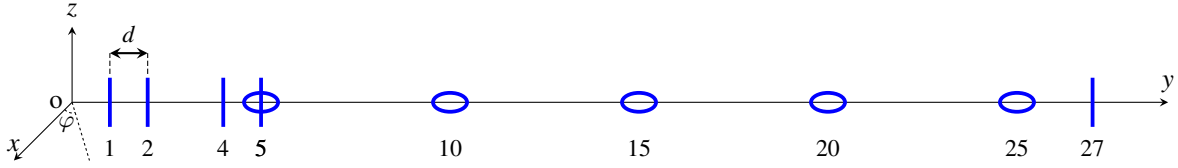


Fig. 2: The structure of SNA with 10 antennas.

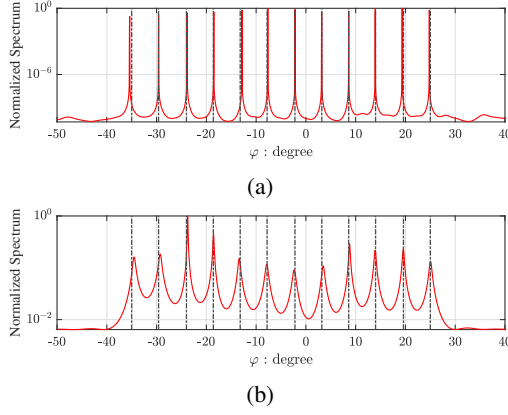


Fig. 3: Spatial spectrum with $K = 12$ obtained by (a) the proposed method, and (b) the two-step method.

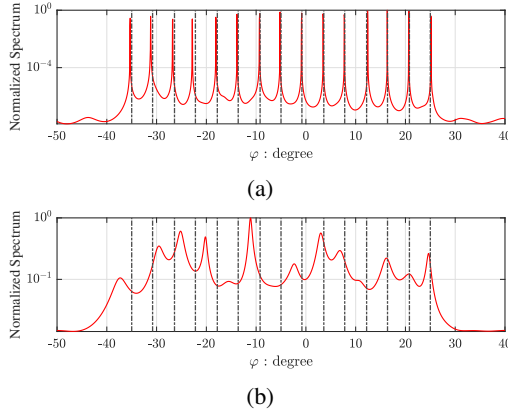


Fig. 4: Spatial Spectrum with $K = 15$ obtained by (a) the proposed method, and (b) the two-step method.

B. Estimation Results

In this subsection, the root mean square error (RMSE) results are examined. The number of sources is set to 13, and the sources are uniformly distributed within $\{(\varphi, \gamma, \eta) \mid \varphi \in [-35^\circ, 45^\circ], \gamma \in [35^\circ, 55^\circ], \eta \in [20^\circ, 70^\circ]\}$. The RMSE results versus SNR with the number of snapshots fixed at 500 are shown in Fig. 5, obtained via 100 Monte Carlo trials. RMSE results versus the number of snapshots with SNR fixed at 0dB are given in Fig. 6. It can be seen from the simulations that the proposed method outperforms the two-step method consistently in the underdetermined scenario.

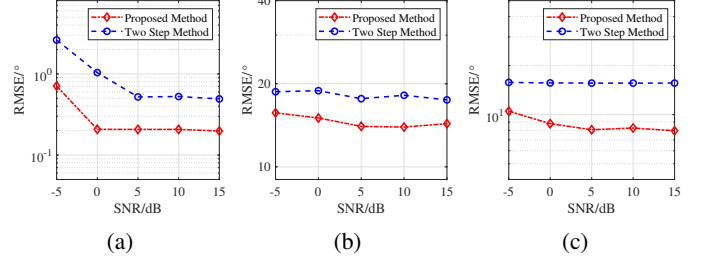


Fig. 5: RMSE results versus SNR. (a) Performance of φ . (b) Performance of γ . (c) Performance of η .

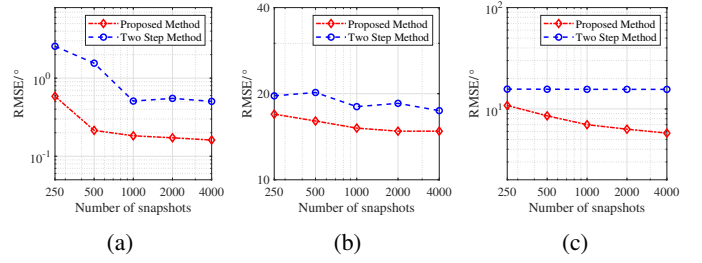


Fig. 6: RMSE results versus the number of snapshots. (a) Performance of φ . (b) Performance of γ . (c) Performance of η .

V. CONCLUSION

An underdetermined DOA and polarization estimation method was proposed in this paper. Based on the virtual array model applicable to arbitrary PSAs, the sparse representation framework was established. Then, DOA estimation exploiting the group sparsity concept was presented, followed by the polarization parameter estimation via the least-square-based method. It has been shown by simulations that based on the same PSA structure, the proposed method is capable of resolving more sources than the existing method, and improved parameter estimation performance can be achieved.

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